

Fig. 2: Pin configurations of 4N33 and BC337

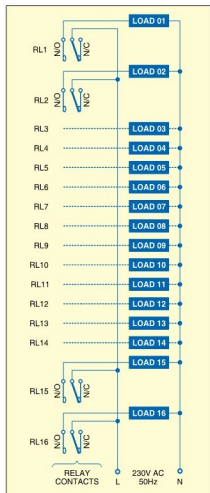


Fig. 3: Load connections

74LS93, 1-of-16 decoder 74LS154, set-reset flip-flop(s) 74LS74, NAND gate(s) 74LS00, regulator 7805 and a few discrete components. AC mains operating frequency is 50 Hz, which is isolated through optocoupler 4N33.

Fig. 2 shows the pin configuration of 4N33 and BC337.

Optocoupler 4N33 consists of a gallium-arsenide infrared LED and

a silicon photo-Darlington transistor. AC mains is connected to pin 1 of 4N33 via current-limiting resistor R3. During the positive half cycle, the internal LED of 4N33 is 'on' and the phototransistor is driven into saturation and

pin 5 goes low. Thus 4N33 provides clock for CD4040 at pin 10.

The CD4040 is a 12-stage ripple-carry binary counter. The counter advances by one count on the negative transition of each clock pulse. It resets to zero with a logical high at the reset input, independent of the clock. Each counter stage is a static toggle flip-flop. Counter CD4040 further divides the 50Hz clock frequency by '10.' Output pin 14 provides clock pulse after an interval of 20.48 seconds and also drives 74LS93.

The 74LS93 is a 4-bit binary ripple counter. It consists of four master-slave flip-flops internally connected to provide a divide-by-two section and a divide-by-eight section. Each section has a separate clock input (CP0 and CP1) to initiate state changes of the counter on the high-to-low clock transition. A gated AND asynchronous master reset (pins 2 and 3) is provided, which resets all the flip-flops.

Since the output from the divide-by-two section is not internally connected to the succeeding stages, in a 4-bit ripple counter the Q0 output must be externally connected to CP1 input. The input count pulses are applied to clock input CP0. Simultaneously, frequency divisions of 2, 4, 8 and 16 are performed at the Q0, Q1, Q2 and Q3 outputs, respectively. The outputs of 74LS93 provide the address inputs to 1-of-16 decoder 74LS154.

The decoder 74LS154 accepts four active-high binary address inputs A0 through A3 and provides 16 mutually exclusive active-low outputs $\bar{O}0$ through $\bar{O}15$. The $\bar{E}0$ and $\bar{E}1$ inputs

PARTS LIST

Semiconductors:

IC1	- 4N33 optocoupler
IC2	- CD4040 divide-by-12 counter
IC3	- 74LS93 binary counter
IC4	- 74LS154, 1-of-16 decoder
IC5-IC12	- 74LS74, D-type flip-flop
IC13	- 74LS00 NAND gate
IC14	- 7805, 5V regulator
T1-T16	- BC337 npn transistor
D1-D20	- 1N4007 rectifier diode
LED1	- 5mm light-emitting diode

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

R1, R4	- 10-kilo-ohm
R2, R3	- 100-kilo-ohm
R5	- 220-ohm
R6-R21	- 1-kilo-ohm

Capacitors:

C1	- 1000 μ F, 35V electrolytic
C2	- 0.1 μ F ceramic disk
C3	- 33 μ F, 16V electrolytic
C4	- 100 μ F, 16V electrolytic

Miscellaneous:

X1	- 230V AC primary to 12V, 500mA secondary transformer
S1	- Push-to-'on' switch
RL1-RL16	- 12V, 1C/O relay

enable the gate, which can be used either to strobe the decoder for eliminating the normal decoding glitches on the outputs, or for expansion of the decoder. The enable gate has two AND'ed inputs which are made low (by connecting them to ground) to enable the outputs. Outputs $\bar{O}0$ through $\bar{O}15$ of 74LS154 are connected to the set inputs of flip-flop 74LS74.

IC5 through IC12 (each 74LS74) are used as set-reset flip-flops to drive the relays with the help of transistors T1 through T16. The 74LS74 is a dual, positive-edge-triggered, D-type flip-flop featuring individual data, clock, set and reset inputs and also true and complementary outputs. Set input 'S' and reset input 'R' are asynchronous active-'low' inputs that operate independently of the clock input. When reset input 'R' is high and set input 'S' is low, the 'Q' output goes high to energise the relay.

The 'Q' outputs of 74LS74 ICs are